p-Laplace Arctan Resonance on the Unit Square

Certified MPA, Shifted-Line RMPA, and JAX + PETSc PMG Results

p-Laplace Arctan Resonance on the Unit Square

Source note used for this implementation: `pLaplace_up_arctan.md` in the repo root.

Mathematical Specification

We study the resonant Dirichlet problem

$$-\Delta_p u = \lambda_1 |u|^{p-2}u + \arctan(u+1) \quad \text{in } \Omega = (0,1)^2, \qquad u = 0 \text{ on } \partial\Omega.$$

with the common nonlinear data

$$g(u) = \arctan(u+1), \qquad g'(u) = \frac{1}{1+(u+1)^2},$$

$$G(t) = \int_0^t g(s)\,ds = (t+1)\arctan(t+1) - \tfrac{1}{2}\log(1+(t+1)^2) - \left(\tfrac{\pi}{4} - \tfrac{1}{2}\log 2\right),$$

so that `G(0)=0`. The energy used by the maintained solver is

$$J_p(u) = \frac{1}{p}\int_\Omega |\nabla u|^p\,dx - \frac{\lambda_1}{p}\int_\Omega |u|^p\,dx - \int_\Omega G(u)\,dx.$$

The common weak form is

$$\int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v\,dx - \lambda_1 \int_\Omega |u|^{p-2}uv\,dx - \int_\Omega \arctan(u+1)v\,dx = 0$$

for all admissible test functions `v`.

p = 2 Validation Problem

$$-\Delta u = 2\pi^2 u + \arctan(u+1) \quad \text{in } (0,1)^2, \qquad u = 0 \text{ on } \partial(0,1)^2.$$

Here the first eigenpair is explicit:

$$\lambda_1 = 2\pi^2, \qquad \varphi_1(x,y) = \sin(\pi x)\sin(\pi y).$$

p = 3 Main Problem

$$-\operatorname{div}(|\nabla u|\nabla u) = \lambda_1 |u|u + \arctan(u+1) \quad \text{in } (0,1)^2, \qquad u = 0 \text{ on } \partial(0,1)^2.$$

The first eigenvalue is not explicit on the unit square when `p=3`, so the workflow first computes a discrete positive eigenpair `(\lambda_{1,h},\varphi_{1,h})` from

$$-\Delta_3 \varphi_1 = \lambda_1 |\varphi_1|\varphi_1, \qquad \varphi_1|_{\partial\Omega} = 0, \qquad \int_\Omega |\varphi_1|^3\,dx = 1.$$

Solvability And Proof Notes

For the source-note specialization of Theorem 6, the key helper quantity is

$$F(x) = \frac{p}{x}\int_0^x g(s)\,ds - g(x), \qquad g(x) = \arctan(x+1).$$

The source note gives the asymptotic limits `lim_{x->+∞} F(x) = (p-1)π/2` and `lim_{x->-∞} F(x) = -(p-1)π/2`. Because `g` is bounded we also have `g(u)/|u|^{p-1} -> 0` as `|u| -> ∞` for both `p=2` and `p=3`. This page therefore claims existence/solvability for both maintained problems. It does not claim global uniqueness, because the source note proves existence, not a uniqueness theorem for the shifted arctan forcing.

Discretization And Algorithm Notes

- Domain: structured `P1` right-triangle meshes on `(0,1)^2`.
- Serial certified mesh ladder: levels `L4`, `L5`, `L6`; level `L7` is reserved for references and eigen diagnostics. The JAX + PETSc extension later in this page continues the same branch to finer levels.
- `p=2` uses the exact `λ₁ = 2π²`.
- `p=3` uses a level-matched discrete `λ₁,h` computed from a cached first-eigenpair stage with `||φ_{1,h}||_{L^3}=1`.
- The maintained solution path is **certified `MPA + stationary Newton`**.
- Raw `MPA/RMPA` runs are diagnostic only. Certified runs use a physically meaningful handoff state and then solve `J'(u)=0` directly.
- The convergence histories report both the coordinate gradient norm and a Laplace-dual finite-element residual norm `||R_h(u)||_{K^{-1}}`.

Certified MPA + Newton

The public results are organized around the discrete gradient, dual residual, and stationary merit

$$g_h(u_h) = \nabla J_{p,h}(u_h), \qquad R_h(u_h) = K^{-1} g_h(u_h), \qquad M(u_h) = \tfrac{1}{2} g_h(u_h)^\top K^{-1} g_h(u_h).$$

The maintained algorithm is:

1. **Raw mountain-pass branch search.** Build a polygonal path from `0` to a positive endpoint `e_h` with `J_{p,h}(e_h)<J_{p,h}(0)`, and at each outer step identify the current path peak `z_k`.
2. **Auxiliary dissertation direction.** Compute the thesis-style descent direction by solving

$$K a_k = g_h(z_k), \qquad d_k = -\frac{a_k}{|a_k|_{1,p,0}}.$$

3. **Best-iterate handoff.** Update the polygonal path by a halved step from the peak, repair the local chain geometry, and retain the iterate with the smallest `\|R_h(z_k)\|_{K^{-1}}` as the certification handoff state.
4. **Certified stationary Newton solve.** Starting from the handoff state, use JAX autodiff to build the discrete gradient and Hessian and solve

$$(H_k + \mu_k K)\,\delta_k = -g_h(u_k).$$

5. **Merit-based globalization.** Accept a trial step `u_{k+1}=u_k+\alpha_k\delta_k` only when

$$M(u_{k+1}) < M(u_k),$$

with regularization and backtracking applied as needed. 6. **Certification stop.** Declare convergence only when

$$\|R_h(u_k)\|_{K^{-1}} \le \varepsilon_{\text{cert}}.$$

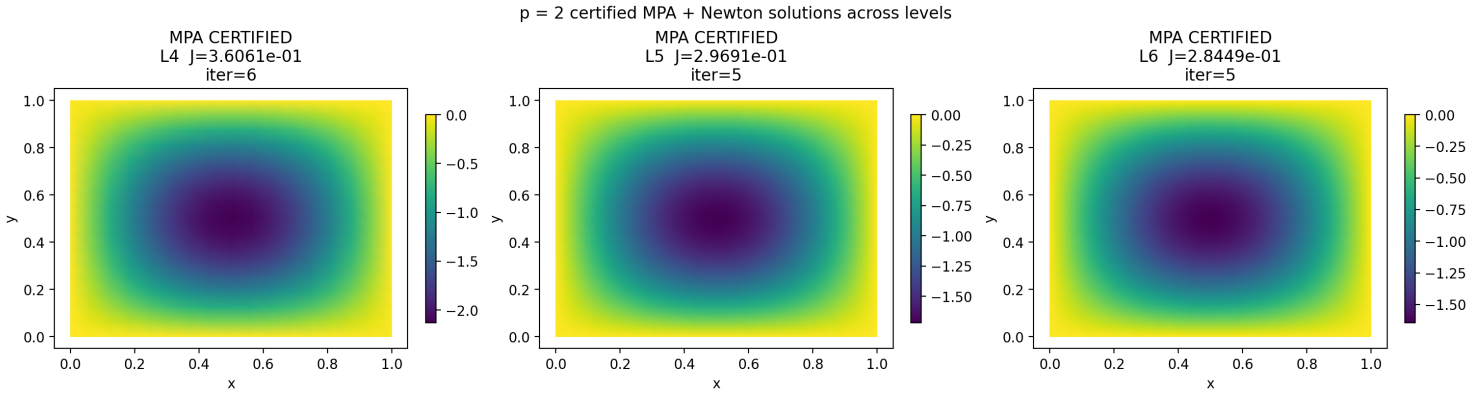
7. **Continuation for `p=3`.** Certify the `p=2` branch first, then continue through `p=2.2, 2.4, 2.6, 2.8, 3.0` on a fixed mesh; each stage uses the previous certified state as the Newton initializer, with raw `MPA` available only as a fallback.

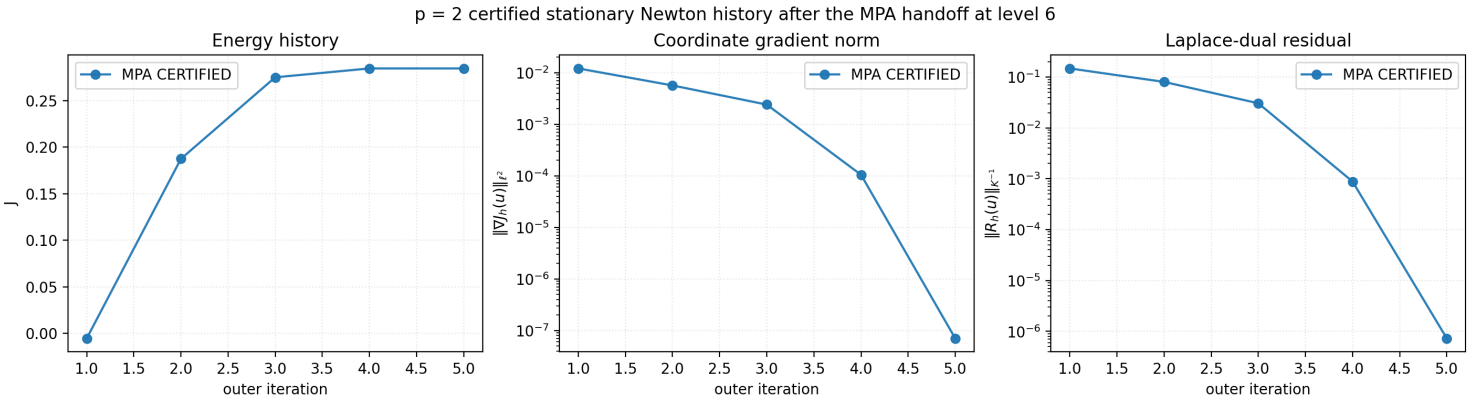
p = 2 Validation Study

The validation problem is reported entirely through the certified workflow. The handoff residual shows the quality of the `MPA` branch-finder state, while the iteration columns show the raw `MPA` work, the Newton work, and their cumulative nonlinear total.

level	certification entry	MPA handoff residual	certified residual	MPA iters	Newton iters	total nonlinear	certified J	stat
4	MPA handoff	0.146882	0.000000	360	6	366	0.360607	com
5	MPA handoff	0.147851	0.000000	360	5	365	0.296914	com
6	MPA handoff	0.148094	0.000000	360	5	365	0.284495	com

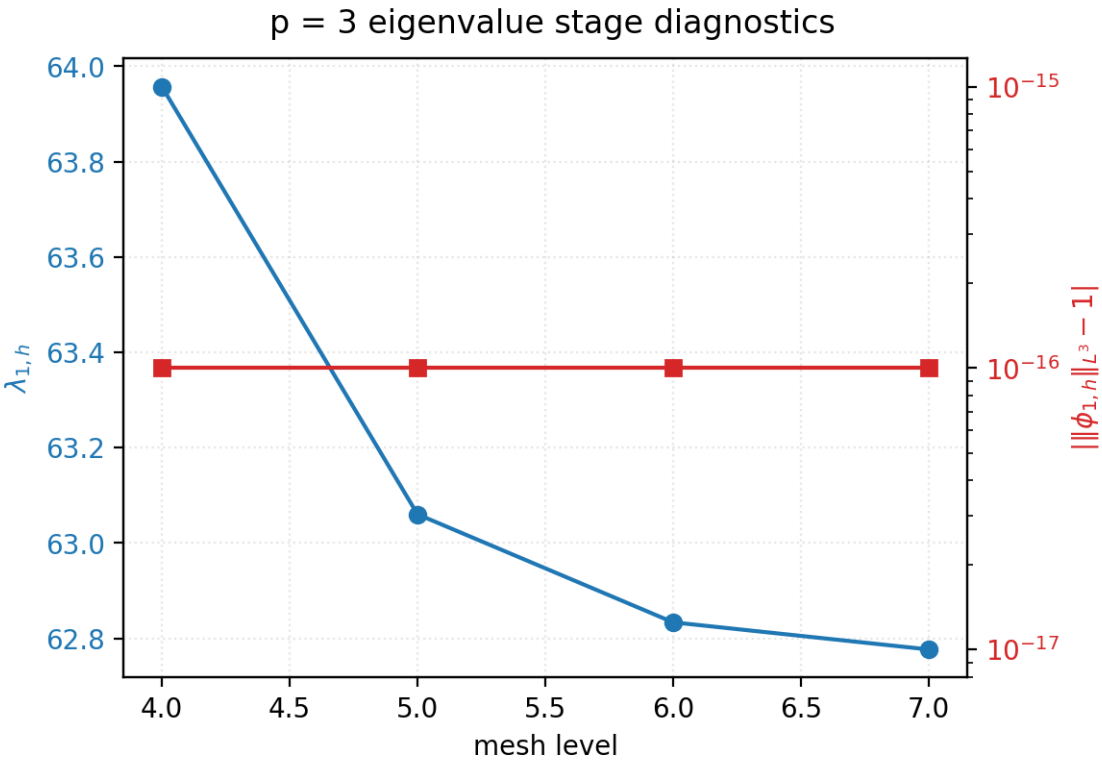
Private `L7` Newton reference residual: `5.823e-07`.

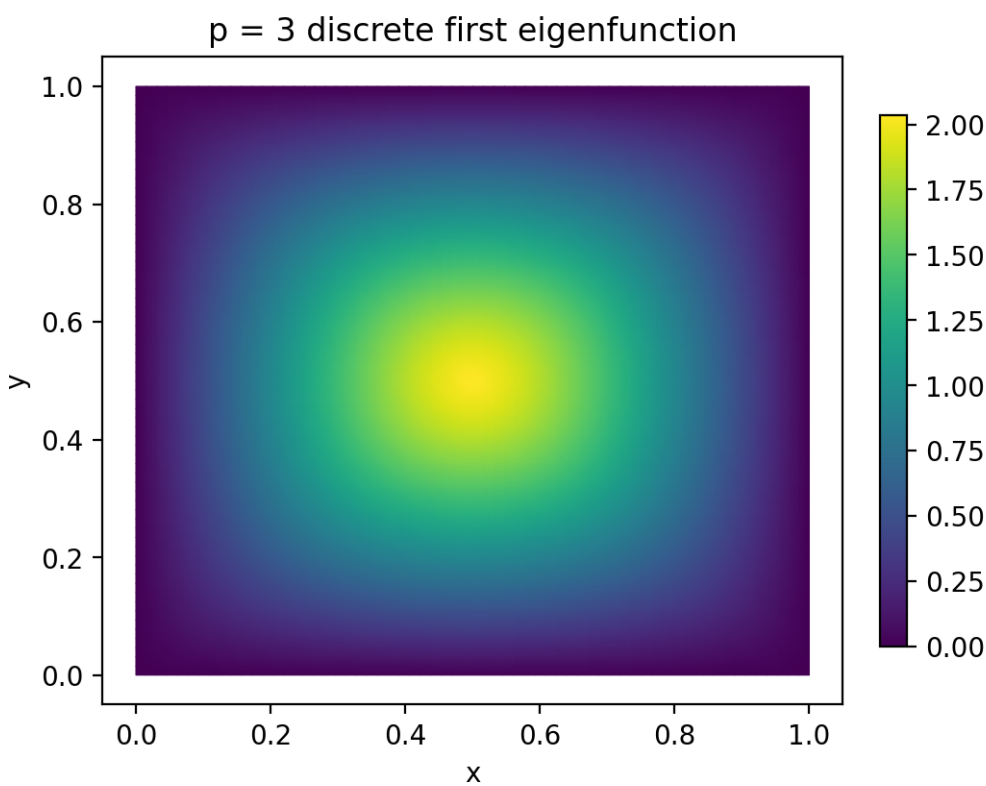




p = 3 Eigenvalue Stage

level	lambda1	residual	norm error	iters	status
4	63.956915	0.000003	0.000000	30	completed
5	63.060087	0.000004	0.000000	30	completed
6	62.833501	0.000005	0.000000	48	completed
7	62.776603	0.000001	0.000000	69	completed





p = 3 Main Study

For `p=3`, the main published solution is the certified continuation branch. The continuation step itself is the essential globalization device; once a certified nearby state is available, the stationary Newton solve is short and robust, and the table reports both the `MPA` contribution and the Newton contribution explicitly.

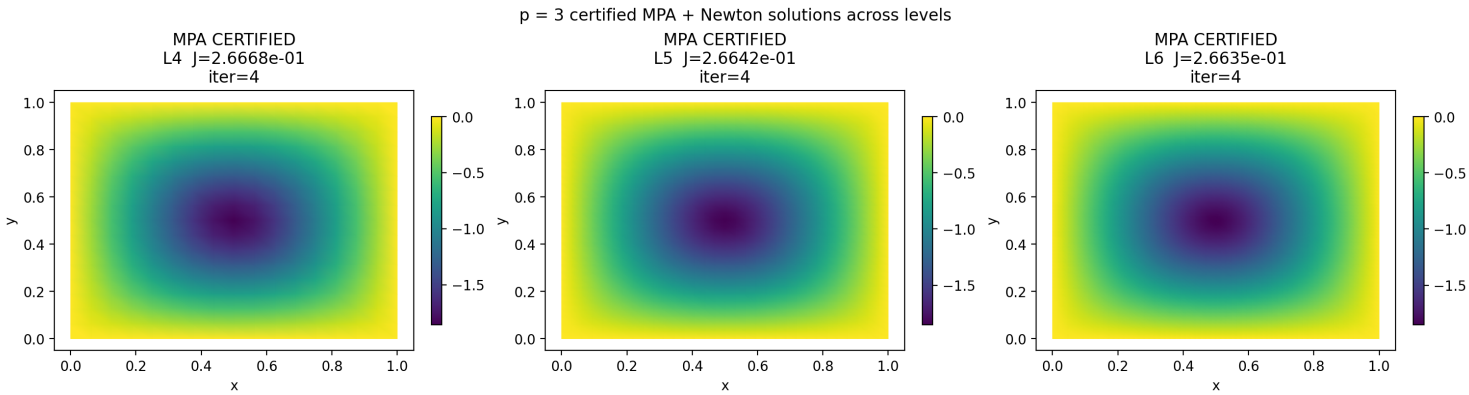
level	lambda1,h	certification entry	certified residual	MPA iters	Newton iters	total nonlinear	certified J	status
4	63.956915	Continuation direct	0.000000	0	4	4	0.266679	completed
5	63.060087	Continuation direct	0.000000	0	4	4	0.266424	completed

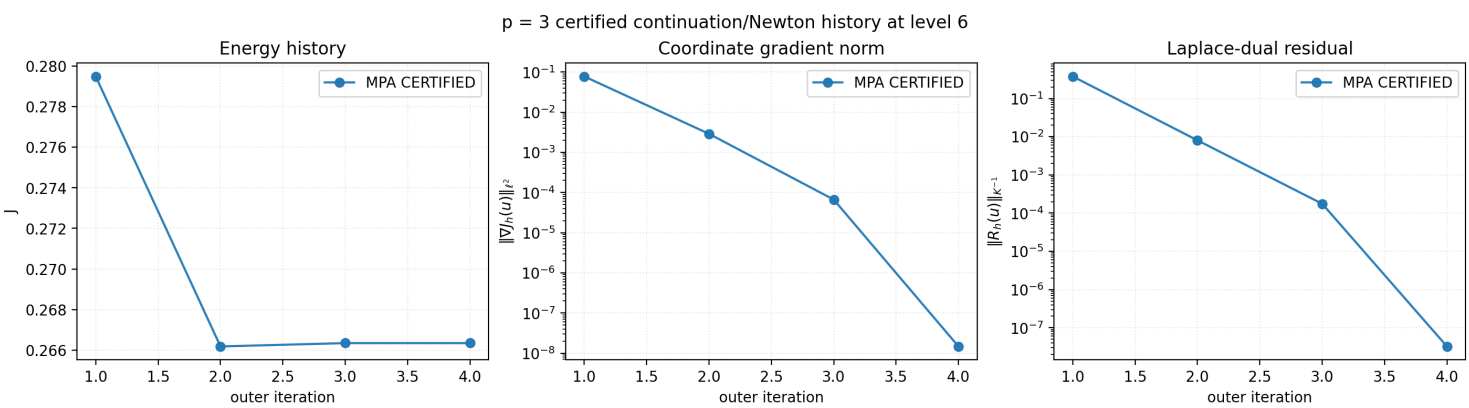
level	lambda1,h	certification entry	certified residual	MPA iters	Newton iters	total nonlinear	certified J	status
6	62.833501	Continuation direct	0.000000	0	4	4	0.266353	completed

Representative `L6` continuation path:

from p	to p	path	certified residual	status
2.0	2.2	direct	0.000000	completed
2.2	2.4	direct	0.000000	completed
2.4	2.6	direct	0.000000	completed
2.6	2.8	direct	0.000000	completed
2.8	3.0	direct	0.000000	completed

Private `L7` certified reference residual: `7.809e-14`.





JAX + PETSc Backend

The local stationary certification solve has now been rewritten as a JAX + PETSc backend. The branch-finding stage is still the certified `MPA` workflow described above, but the expensive fine-mesh Newton solve now uses PETSc `FGMRES` with a structured PMG preconditioner built from the stiffness matrix.

The Hessian assembly uses a zero-safe element regularization

$$|x|^q_{\varepsilon_h} = \left(x^2 + \varepsilon_h^2\right)^{q/2} - \varepsilon_h^q, \qquad \varepsilon_h = 10^{-12},$$

which keeps the transferred warm-start Hessians finite while remaining visually indistinguishable from the unsmoothed functional at plotting scale.

Accepted PMG configuration

- Krylov solver: `FGMRES`
- Preconditioner: fixed-stiffness Galerkin PMG
- Multigrid smoother: **Chebyshev + Jacobi**
- Coarse level: PETSc default coarse backend from the solver CLI

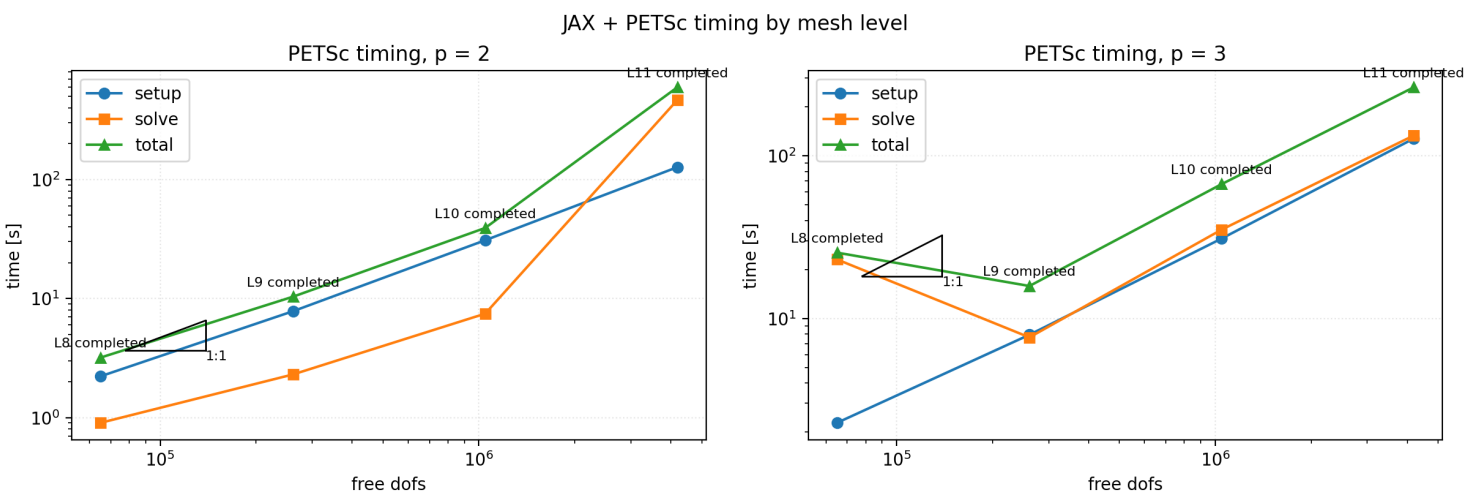
- Near the tolerance floor: a small stagnation guard accepts convergence when the residual is already at the target scale and no further regularized step can produce a numerically resolvable merit decrease

PETSc Timing And Scaling

The PETSc timing study extends the maintained ladder beyond the serial publication meshes. For `p=2` the warm-started PMG solve is continued to finer levels, and for `p=3` the PETSc backend is reported as a tuned continuation backend using the finest available certified eigenvalue cache from the serial study. The PETSc tables expose nonlinear outer iterations, Newton iterations, and cumulative Krylov iterations so the PMG workload is visible directly in the published summary. In this backend the nonlinear counter coincides with the Newton outer loop. The timing figure is log-log with an ideal `1:1` triangle, and the strong-scaling speedup panel is also shown on log-log axes against the ideal line.

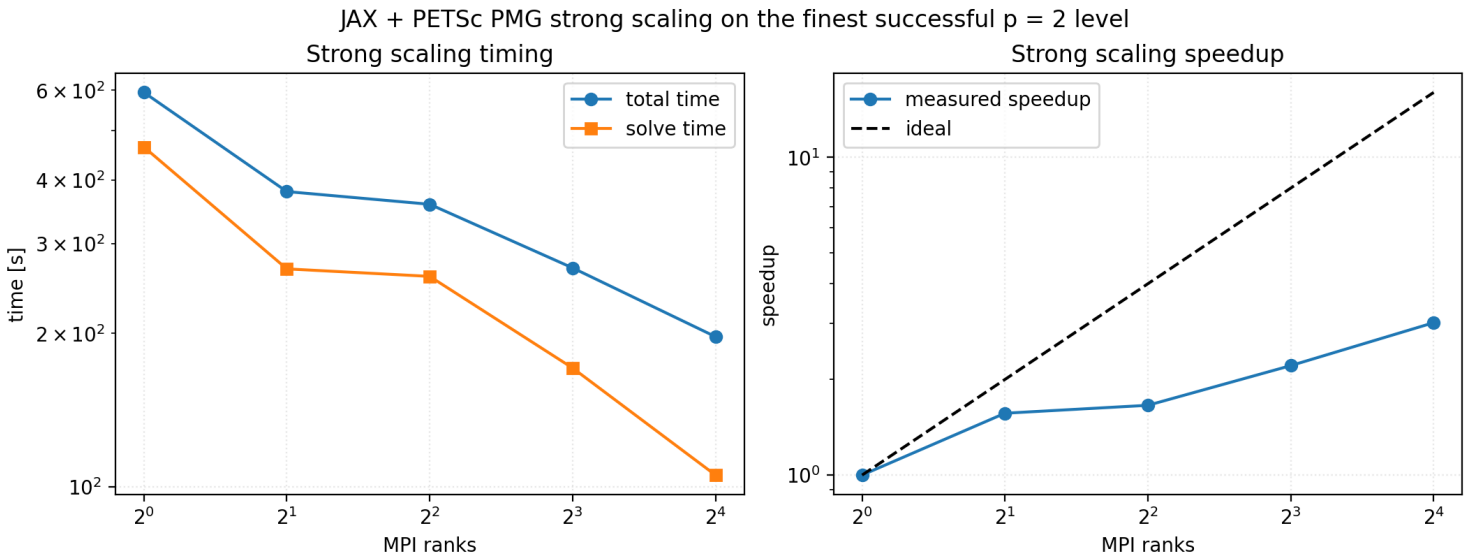
level	free dofs	status	residual	setup [s]	solve [s]	total [s]	nonlinear its	Newton iters	linear its
8	65025	completed	0.000000	2.222	0.903	3.192	3	3	42
9	261121	completed	0.000000	7.824	2.303	10.361	3	3	44
10	1046529	completed	0.000000	30.873	7.463	39.167	3	3	47
11	4190209	completed	0.000000	126.844	464.449	594.462	6	6	119

level	free dofs	status	residual	setup [s]	solve [s]	total [s]	nonlinear its	Newton iters	linear its
8	65025	completed	0.000000	2.279	23.035	25.375	7	7	571
9	261121	completed	0.000000	7.930	7.642	15.817	4	4	462
10	1046529	completed	0.000000	30.970	35.139	66.922	4	4	603
11	4190209	completed	0.000000	127.456	133.156	263.536	3	3	609



Strong-scaling rows are reported on the finest successful `p=2` PETSc level `L11`.

ranks	status	total [s]	nonlinear its	Newton iters	linear its	speedup	efficiency	residual
1	completed	594.462	6	6	119	1.000	1.000	0.000000
2	completed	379.720	5	5	84	1.566	0.783	0.000000
4	completed	358.531	8	8	129	1.658	0.415	0.000000
8	completed	268.941	7	7	110	2.210	0.276	0.000000
16	completed	197.076	5	5	77	3.016	0.189	0.000000



Alternative Certified Branch: Shifted-Line RMPA + Newton

A second certified branch-finding variant is now documented at the representative mesh level `L6`. Instead of projecting onto the origin-centred ray `t w`, it fixes a negative anchor `-C_1\phi_h`, numerically maximizes the energy along the affine line through that anchor, and then uses the same stationary Newton certification stage. This is still presented as an alternative branch finder rather than the maintained headline workflow, because its success is more seed-sensitive than the continuation-guided certified `MPA + Newton` path. The point of the section is to document the geometry change and its measured effect, not to replace the maintained solver story above.

p	best seed	raw status	raw residual	raw its	certified status	certified residual	Newton iters	time [s]
2	sine	completed	0.000001	9	completed	0.000000	1	2.655
3	eigenfunction	completed	0.000004	22	completed	0.000000	1	7.252

Seed And Endpoint Geometry Comparison

These tables keep the mesh fixed at `L6` so only the branch-search geometry and the start seed vary. For the MPA family, the geometry axis is one-sided `0 -> +C\phi_h` versus symmetric `-C_1\phi_h -> +C_2\phi_h`. For the RMPA family, the geometry axis is the classical ray from `0` versus the shifted line from `-C_1\phi_h`.

MPA Family

p	geometry	seed	raw status	raw residual	raw its	certified status	certified residual	Newton iters
2	0 -> +C seed	sine	maxit	0.148094	360	completed	0.000000	5
2	0 -> +C seed	bubble	maxit	0.148094	360	completed	0.000000	5
2	0 -> +C seed	tilted	maxit	0.148094	360	completed	0.000000	5
2	-C1 seed -> +C2 seed	sine	maxit	0.020512	360	completed	0.000000	2
2	-C1 seed -> +C2 seed	bubble	maxit	0.020512	360	completed	0.000000	2
2	-C1 seed -> +C2 seed	tilted	maxit	0.020512	360	completed	0.000000	2
3	0 -> +C seed	sine	maxit	0.136651	360	failed	0.113560	30
3	0 -> +C seed	bubble	maxit	0.136651	360	failed	0.113560	30
3	0 -> +C seed	tilted	maxit	0.136651	360	failed	0.113560	30
3	0 -> +C seed	eigenfunction	maxit	0.136651	360	failed	0.113560	30
3	-C1 seed -> +C2 seed	sine	maxit	0.597809	360	completed	0.000000	6
3	-C1 seed -> +C2 seed	bubble	maxit	0.597809	360	completed	0.000000	6
3	-C1 seed -> +C2 seed	tilted	maxit	0.597809	360	completed	0.000000	6
3	-C1 seed -> +C2 seed	eigenfunction	maxit	0.597809	360	completed	0.000000	6

RMPA Family

p	geometry	seed	raw status	raw residual	raw its	certified status	certified residual	Newton iters	tot
2	ray from 0	sine	failed	0.185979	0	completed	0.000000	6	6
2	ray from 0	bubble	failed	0.185979	0	completed	0.000000	6	6
2	ray from 0	tilted	failed	0.185979	0	completed	0.000000	6	6
2	line from -C1 seed	sine	completed	0.000001	9	completed	0.000000	1	10
2	line from -C1 seed	bubble	completed	0.000004	9	completed	0.000000	1	10
2	line from -C1 seed	tilted	completed	0.000004	8	completed	0.000000	1	9
3	ray from 0	sine	failed	1.209430	0	failed	0.167461	30	30
3	ray from 0	bubble	failed	1.209430	0	failed	0.167461	30	30
3	ray from 0	tilted	failed	1.209430	0	failed	0.167461	30	30
3	ray from 0	eigenfunction	failed	1.209430	0	failed	0.167461	30	30
3	line from -C1 seed	sine	failed	1.209430	0	failed	0.167461	30	30
3	line from -C1 seed	bubble	failed	2.603957	0	failed	0.201878	38	38
3	line from -C1 seed	tilted	failed	1.665968	0	completed	0.000000	8	8
3	line from -C1 seed	eigenfunction	completed	0.000004	22	completed	0.000000	1	23

Cross-Method Comparison

The cross-method material is retained for completeness, but it is annexed below so the main narrative stays focused on the certified `MPA + stationary Newton` path.

Raw Versus Certified Diagnostics

The remainder of the page is annex material. It records the raw globalization behavior and the failing `RMPA` variants so the successful certified path above stays readable and unambiguous.

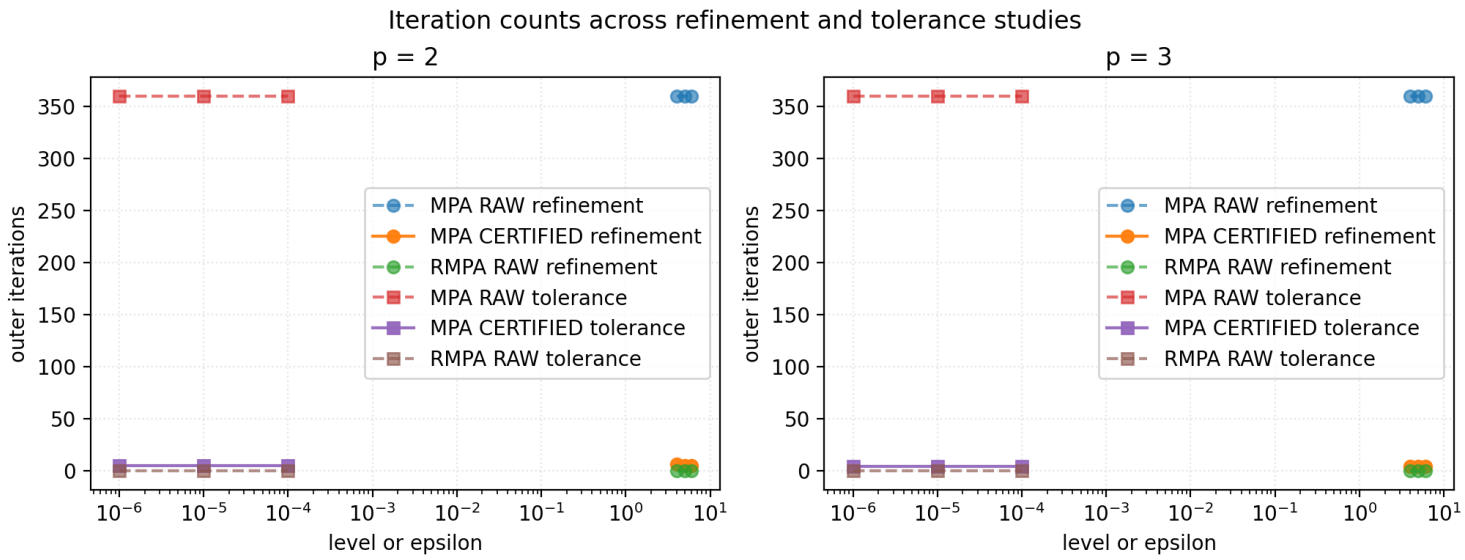
Annex A — Cross-Method Diagnostics

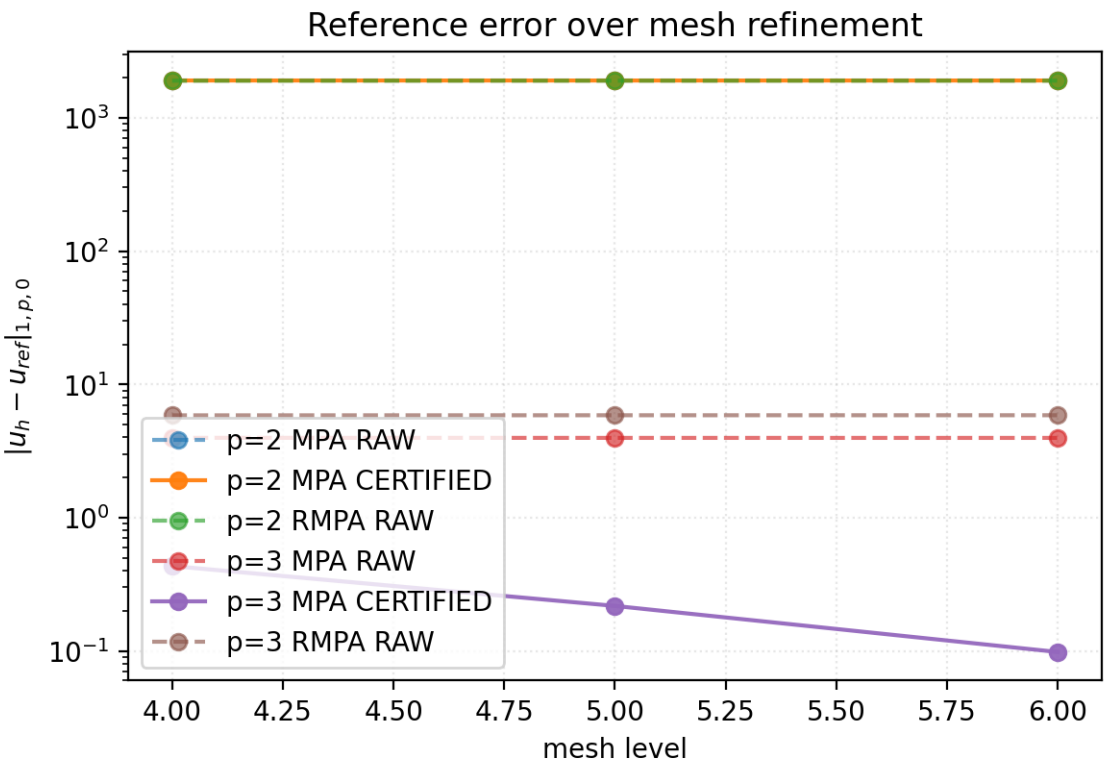
`p=2` mesh-refinement diagnostics

method	level	raw status	raw residual	certified status	certified residual	raw seed	certified handoff
MPA	4	maxit	0.146882	completed	0.000000	sine	MPA handoff
MPA	5	maxit	0.147851	completed	0.000000	sine	MPA handoff
MPA	6	maxit	0.148094	completed	0.000000	sine	MPA handoff
RMPA	4	failed	0.165378	failed	0.165378	sine	-
RMPA	5	failed	0.181815	failed	0.181815	sine	-
RMPA	6	failed	0.185979	failed	0.185979	sine	-

`p=3` mesh-refinement diagnostics

method	level	raw status	raw residual	certified status	certified residual	raw seed	certified handoff
MPA	4	maxit	0.136526	completed	0.000000	eigenfunction	Continuation direct
MPA	5	maxit	0.136630	completed	0.000000	eigenfunction	Continuation direct
MPA	6	maxit	0.136651	completed	0.000000	eigenfunction	Continuation direct
RMPA	4	failed	1.188997	failed	1.188997	eigenfunction	-
RMPA	5	failed	1.205242	failed	1.205242	eigenfunction	-
RMPA	6	failed	1.209430	failed	1.209430	eigenfunction	-





Annex B — RMPA And Failed Paths

p=2 RMPA diagnostics

level	raw status	raw residual	ray kind	stable interior ray maximum?	rationale
4	failed	0.165378	maximum	no	Certification skipped because the ray audit did not sh
5	failed	0.181815	maximum	no	Certification skipped because the ray audit did not sh
6	failed	0.185979	maximum	no	Certification skipped because the ray audit did not sh

p=3 RMPA diagnostics

level	raw status	raw residual	ray kind	stable interior ray maximum?	rationale
4	failed	1.188997	maximum	no	Certification skipped because the ray audit did not sh
5	failed	1.205242	maximum	no	Certification skipped because the ray audit did not sh
6	failed	1.209430	maximum	no	Certification skipped because the ray audit did not sh

Annex C — Tolerance Comparison

p	method	epsilon	raw status	raw residual	certified status	certified residual
2	MPA	1e-04	maxit	0.148094	completed	0.000000
2	MPA	1e-05	maxit	0.148094	completed	0.000000
2	MPA	1e-06	maxit	0.148094	completed	0.000000
2	RMPA	1e-04	failed	0.185979	failed	0.185979
2	RMPA	1e-05	failed	0.185979	failed	0.185979
2	RMPA	1e-06	failed	0.185979	failed	0.185979
3	MPA	1e-04	maxit	0.136651	completed	0.000000
3	MPA	1e-05	maxit	0.136651	completed	0.000000
3	MPA	1e-06	maxit	0.136651	completed	0.000000
3	RMPA	1e-04	failed	1.209430	failed	1.209430
3	RMPA	1e-05	failed	1.209430	failed	1.209430
3	RMPA	1e-06	failed	1.209430	failed	1.209430

Why `RMPA` stays in the annexes:

- The classical origin-based `RMPA` still does not show the stable interior ray maximum required by its projection logic on the positive arctan branch.
- The shifted-line variant documented above can work well, but only for some seed and geometry combinations, so it is still presented as a secondary certified branch finder rather than the maintained headline method.
- Tightening the raw tolerance does not materially fix the origin-based `RMPA` rows on the published ladder.
- The maintained successful path remains **certified `MPA + stationary Newton`**, with continuation in `p` providing the decisive stabilization for `p=3`.

Commands Used

```
./.venv/bin/python -u experiments/runners/run_plaplace_up_arctan_suite.py \
  --out-dir artifacts/raw_results/plaplace_up_arctan_full \
  --summary artifacts/raw_results/plaplace_up_arctan_full/summary.json
```

```
./.venv/bin/python -u experiments/runners/run_plaplace_up_arctan_petsc_suite.py \
  --out-dir artifacts/raw_results/plaplace_up_arctan_petsc \
  --summary artifacts/raw_results/plaplace_up_arctan_petsc/summary.json
```

```
./.venv/bin/python -u experiments/analysis/generate_plaplace_up_arctan_report.py \
  --summary artifacts/raw_results/plaplace_up_arctan_full/summary.json \
  --petsc-summary artifacts/raw_results/plaplace_up_arctan_petsc/summary.json \
  --out artifacts/reports/plaplace_up_arctan/README.md \
  --asset-dir docs/assets/plaplace_up_arctan
```

```
./.venv/bin/python -u experiments/analysis/generate_plaplace_up_arctan_problem_page.py \
  --summary artifacts/raw_results/plaplace_up_arctan_full/summary.json \
  --petsc-summary artifacts/raw_results/plaplace_up_arctan_petsc/summary.json \
  --out docs/problems/pLaplace_up_arctan.md \
  --asset-dir docs/assets/plaplace_up_arctan
```

Artifacts

- Raw study summary: `artifacts/raw_results/plaplace_up_arctan_full/summary.json`
- JAX + PETSc timing summary: `artifacts/raw_results/plaplace_up_arctan_petsc/summary.json`
- Internal report: `artifacts/reports/plaplace_up_arctan/README.md`
- Cached `p=3` eigen stage: `artifacts/raw_results/plaplace_up_arctan_full/lambda_cache/`
- No separate `docs/results/pLaplace_up_arctan.md` page is maintained for this family.